# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** In the Sierpinski triangle construction, the number of small shaded triangles at stage 1 (stage 0 has one) is: 1

**(a)** 3 **(b)** 6 **(c)** 2 **(d)** 4

**2.** The number of faces of a triangular prism is: 1

**(a)** 3 **(b)** 5 **(c)** 4 **(d)** 6

**3.** The number of vertices of a triangular pyramid is: 1

**(a)** 6 **(b)** 3 **(c)** 5 **(d)** 4

**4.** The number of edges of a triangular prism is: 1

**(a)** 8 **(b)** 9 **(c)** 6 **(d)** 5

**5.** A polyhedron has 6 faces and 8 vertices. By Euler's formula, its number of edges is: 1

**(a)** 14 **(b)** 24 **(c)** 13 **(d)** 12

**6.** The sum of the lengths of all the edges of a cube of side 3 cm is (in cm): 1

**(a)** 12 **(b)** 36 **(c)** 24 **(d)** 18

**7.** Euler's formula for a polyhedron with $F$ faces, $V$ vertices and $E$ edges is: 1

**(a)** $F - V + E = 2$ **(b)** $F + V = E$ **(c)** $F + V + E = 2$ **(d)** $F + V - E = 2$

**8.** In the Sierpinski triangle construction, the number of small shaded triangles at stage 2 (stage 0 has one) is: 1

**(a)** 12 **(b)** 8 **(c)** 6 **(d)** 9

**9.** The number of faces of a pentagonal prism is: 1

**(a)** 7 **(b)** 6 **(c)** 5 **(d)** 10

**10.** The number of vertices of a pentagonal pyramid is: 1

**(a)** 7 **(b)** 6 **(c)** 5 **(d)** 10

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Write the number of faces, vertices and edges of a triangular prism. 2

**12.** A solid has 6 faces, 8 vertices and 12 edges. Verify Euler's formula. 2

**13.** Find the total length of all edges of a cube of side 4 cm. 2

**14.** Write the number of faces, vertices and edges of a pentagonal prism. 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** State the number of faces, vertices and edges of a triangular pyramid and verify Euler's formula. 3

**16.** In the Sierpinski triangle, how many shaded triangles are there at stage 2? How many more are there at stage 3? 3

**17.** A cuboid measures 5 cm by 3 cm by 2 cm. Find the total length of its edges and state its number of faces and vertices. 3

OR

State the number of faces, vertices and edges of a square pyramid and verify Euler's formula. 3

**18.** In the Sierpinski triangle, how many shaded triangles are there at stage 3? How many more are there at stage 4? 3

OR

A cuboid measures 8 cm by 4 cm by 3 cm. Find the total length of its edges and state its number of faces and vertices. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A Koch curve begins with a straight line of length 9 cm (stage 0). At each stage every segment is replaced by 4 segments each one-third as long. Find the number of segments, the length of each segment and the total length at stage 2. 5

OR

In the Sierpinski triangle, stage 0 has 1 shaded triangle. Fill in the number of shaded triangles at stages 1, 2 and 3. By what factor does the count grow at each stage? 5

**20.** A Koch curve begins with a straight line of length 18 cm (stage 0). At each stage every segment is replaced by 4 segments each one-third as long. Find the number of segments, the length of each segment and the total length at stage 2. 5

OR

In the Sierpinski triangle, stage 0 has 1 shaded triangle. Fill in the number of shaded triangles at stages 2, 3 and 4. By what factor does the count grow at each stage? 5

## END OF QUESTION PAPER Total: 40 Marks

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